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MIL-P-23108(Wep)
30 NOVEMBER 1961

DOWNGRADED BY AUTHORITY OF
NAVORDINST 5511.35 ORD-065
DATED 7 MAY 1973

MILITARY SPECIFICATION

PLUG IN UNIT FOR TEST SET, ECHO SIMULATOR (U)

(Mk 315 Mod 1 for use with Torpedo Mark 32)

This specification has been approved by the Bureau of Naval Weapons
Department of the Navy

1. SCOPE

1.1 Scope. -This specification establishes the requirements for the procurement of a Plug In Unit, Bureau of Naval Weapons LD 512143 which when used with the Main Unit Assembly (Test Set Echo Simulator Mk 315 Mod 1) can provide signals to and monitor the performance of the Torpedo Mk 32 Mod 2 Acoustic Panels. The Unit is of one type and class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation forbids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901 (Navy)	Shockproof Equipment Class HI (High Impact), Shipboard Application, Tests for
MIL-T-945	Test Equipment for use with Electronic Equipment: General Specification
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of

DOWNGRADED AT 12 YEAR
INTERVALS NOT AUTOMATICALLY
DECLASSIFIED. DOD DIR 5200.10

FSC 1355

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MIL-W-21927	Weapons; Handling and Preparation for Delivery of; General Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 512143	Plug-In Unit Mk 32 Assembly

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OP 400	General Instructions for Design, Manufacture and Inspection of Naval Ordnance Equipment
OP 2085	Torpedo Mark 32 Mod 2, Description, Operation and Maintenance
NAVORD OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.- The following documents form a part of this specification to the extent specified herein, Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

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Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Description.- The Plug In Unit Mk 32 is an electronic assembly which is used with the Basic Unit of Test Set Mark 315 Mod 1 to simulate target signals for the Torpedo Mk 32. An external or internal pulse triggers a multi-vibrator circuit in which variable time delay is introduced by the Plug In Unit. The delayed pulse then triggers a second multi-vibrator circuit and the Plug In Unit establishes the pulse length. This signal gates an oscillator whose output is converted into selective signals compatible with the torpedo being tested.

3.2 General requirements

3.2.1 Preproduction units. Unless otherwise specified in the contract or order, preproduction units of the plug in unit shall be manufactured using the methods and procedures proposed for production. These units will be tested as specified in section 4 and are for the purpose of determining that the production methods are capable of yielding items that comply with the technical requirements of the contract.(.See 4.2.1)

3.2.2 Materials.- Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the unit shall conform to the applicable Federal or Military Specifications. Materials which are not covered by specification or which are not specifically designated herein shall be of the best quality of the lightest practicable weight consistent with strength and durability, and entirely suited to the purpose for which intended under all conditions of test and service operation. Recommendations concerning the use of a superior substitute for any part or material specified herein shall be accompanied by a description of the proposed substitute, the reasons therefore, together with evidence to substantiate any claim as to its suitability. At the discretion of the Bureau of Naval Weapons, test samples may be required to prove the suitability of the proposed substitute.

3.2.3 Surface treatment, construction and dimensions.- Surface treatment, construction and dimensions for the Unit shall be in accordance with the applicable drawings, specifications, and publications listed in BuWeps LD 512143.

3.2,4 Moisture, fungus and corrosion. - The Unit shall be rendered moisture, fungus and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in BuWeps LD 512143, When moisture, fungus and corrosion resistance are not specified in these documents, the item shall conform as required by NavOrd OS 6341.

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3.2.5 Marking.- Marking of assemblies, subassemblies and parts shall be in accordance with the applicable drawings listed in BuWep LD 512143.

3.2.6 Interchangeability.- Individual parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with mating parts of the unit and other like units. If deviations greater or smaller than the tolerances shown on the drawings or specifications are required for interchangeability; or if hand fitting beyond that shown on the applicable drawings is required; then changes in dimensions are necessary and revision and correction of the applicable drawings or specification is required. (See 3.2.9.)

3.2.7 Threads.- Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.2.8 Requests for deviation.- Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.2.9 Conflicting requirements.- Conflicting requirements arising between this specification or any specification, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification and resolution (or correction).

3.3 Definitions.- Definitions applicable to the environmental requirements of this specification are as follows.

3.3.1 Nonoperating condition. - The condition of the unit when stored, transported, and handled.

3.3.2 Ready, nonoperating condition.- The condition of the u nit when issued, transported, and handled.

3.3.3 Vibration.- Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement is given in inches peak to peak or the corresponding maximum acceleration in gravitational units (g). The frequency is given in cycles per second (cps).

3.3.4 Shock.- Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a build-up of 90 percent of peak force in less than 0.010 sec. Shock is described (a) as a pulse in g, duration in milliseconds (ins) and some description of the shock wave form, or (b) in terms of test conditions outlined in Specification MIL-S-901.

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3.3.5 Unit axes. -The axes of the unit are:

(a) Longitudinal

(b) Perpendicular and transverse to (a)

(c) Mutually perpendicular to (a) and (b).

3.4 Performance requirements and product characteristics.- The Plug In Unit, when operated with the Basic Unit, shall meet the following performance requirements and product characteristics. (There is no satisfactory way in which to evaluate the Plug In Unit other than in conjunction with the Basic Unit or with a piece of test equipment possessing all of the characteristics of the Basic Unit).

3.4.1 Physical compatibility.- The Plug In Unit must be physically compatible for insertion, securing, and removal from the Basic Unit.

3.4.2 Electrical compatibility.- The Plug In Unit shall meet all of the perform-ante requirements listed herein when supplied with the voltages from a properly operating Basic Unit,

3.4.3 Input signal.- The unit shall respond to input trigger pulses having a frequency of 60 ± 2 kilocycles (kc), peak to peak amplitude of 760 ± 40 volts, pulse length of two plus or minus 0.5 milliseconds (ins) and a pulse repetition rate of 96 ± 6 pulses per minute.

3.4.4 Output echo signal.- The response to an input signal as per 3,4,3 shall be a generated echo signal output pulse with the following characteristics.

3.4.4.1 Pulse frequency.- The frequency of the echo signal pulse shall be equal to the frequency of the internal crystal used and shall be capable of being changed from 60.0. to 60.4 kc by means of the Frequency Select Switch on the panel face.

3.4.4.2 Time delay (range).- The time delay from the leading edge of an input trigger pulse (3.4.3) to the leading edge of the echo signal pulse shall be capable of being switched to the following fixed delay values: (a) 80 milliseconds corresponding to 200 feet; and (b) 480 milliseconds corresponding to 1200 feet.

3.4.4.3 Pulse length.- The pulse length of the echo signal pulse shall be capable of being adjusted to 35 milliseconds ± 20 percent.

3.4.4.4 Continuous wave (CW) output voltage.- The CW output voltage shall be as follows when measured between pins E and D (ground) of J10 with the zero-down up switch in the UP position.

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Attenuation (db)	Output Voltage (rms)
0	10,000 microvolt ± 5%
20	1,000 microvolts ± 5%
40	100 microvolt ± 5%
60	10 microvolt ± 5%

3.4.4.5 Noise level.- With the unit set for pulse operation and with no input signal (3.4.3) applied, the noise level of the CW output voltage (3.4.4.4) shall be not greater than ten microvolt (uv) root mean square (rms) at crystal frequency.

3.4.4.6 Wave shape.- The wave shape of the echo signal shall have a maximum rise and decay time of six cycles. The overshoot shall not exceed 15 percent of the required magnitude.

3.4.5 High level output.- The unit shall be capable of providing a separate CW high level output signal having a no load voltage magnitude of at least seven volts rms and at a frequency of the crystal used.

3.4.6 Manual operation. - A manual trigger switch shall cause the unit to respond as per 3.4.4 without the application of an input signal as per 3.4.3.

3.4.7 Performance requirements and product characteristics to be tested for only in preproduction and periodic tests.

3.4.7.1 Torpedo Mark 32 Mod 2 Control Panel Checks.- The Plug In Unit, in conjunction with the Basic Unit, shall be capable of performing the Mk 32 Mod 2 Control Panel checks as specified in OP 2085.

3.5 Environmental requirements.- The unit shall be capable of withstanding the environmental requirements and tests of this specification and thereafter meet all the performance requirements and product characteristics specified herein.

3.5.1 Vibration, nonoperating condition.- The unit shall withstand vibrations of 3 g over the frequency range ten to 100 cps. These vibrations may occur in any direction. Thereafter the unit shall show no damage and shall meet all the performance requirements and product characteristic of this specification.

3.5.2 Shock, ready, nonoperating.- The unit shall receive a shock in accordance with the instructions as outlined in Military Specification MIL-T-945. Thereafter the unit shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.5.3 Shipping condition

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3.5.3.1 Shock-transportation, storage and handling.- The unit after being subjected to the tests specified in 4.5.5.3.1, shall show no damage and shall meet all the requirements of section 3.

3.5.3.2 Vibration-transportation.- The panel, after being subjected to the tests specified in 4.5.5.3.2, shall show no damage and shall meet all the requirements of section 3.

3.5.4 Temperature, ready or operating condition.- The unit shall meet the performance requirements and product characteristics of this specification while operating at temperatures between 30 and 130 degrees Fahrenheit (F).

3.5.5 Temperature; ready nonoperating. - The unit shall be capable of being maintained in a nonoperating condition for extended periods in temperatures within the range of minus 40 degrees F to 160 degrees F. When returned to the operating temperature specified in 3.5.4 the unit shall meet the performance requirements of this specification.

3.5.6 Operating life.- The unit shall have a continuous operating life of at least 720 hours during which the unit shall be triggered at the rate of at least one pulse per second. At the end of each 24 hour period the unit shall be capable of meeting the performance requirements and product characteristics of this specification.

3.5.7 Humidity.- With the unit in a ready nonoperating condition and plugged into a main unit assembly (Test Set Echo Simulator Mk 315 Mod 1), place the complete unit in a temperature humidity chamber. Maintain the humidity between 85 and 95 percent during the following temperature cycles; Raise the temperature to 130 degrees F' and maintain this condition for one hour, lower the temperature to 90 degrees F. The unit shall meet the performance requirements of this specification. The rate of temperature change shall not be less than ten degrees per hour,

3.6 Clean-up.- Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship.- The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, riveting, machine screw assemblies, welding, brazing and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

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4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Gages.- The contractor shall provide himself with whatever gages are necessary and adequate to insure that the materials to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

4.1.2 Preproduction units. Unless otherwise specified for each contract or order, five plug in units shall be manufactured using the same methods and procedures proposed for the production. These shall be submitted to the Government inspector for preproduction testing at an activity designated in the contract or by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, drawings, and specifications. Further production of the item by the contractor prior to the completion of tests on the preproduction units shall be at the contractor's risk. Preproduction units accepted will be applied as part of the quantity specified by the contract or order. Preproduction tests are detailed in 4.5. Acceptance shall be based on no defects in the unit.

4.1.3 Lot size. Ail units offered for delivery at one time shall be considered a lot for purposes of inspection.

4.2 Sampling. - When tests and examinations are deemed necessary to verify product acceptability, the Government representative shall use sampling plans and procedures outlined in MIL-STD-105.

4.3 Classification of test.- The inspection and testing of the unit shall be classified as follows.

4.3.1 Acceptance tests.- Unless otherwise specified, these tests are to be accomplished by the contractor on each unit being submitted for acceptance under contract, These tests are detailed in 4.5.

4.3.2 Preproduction tests.- Preproduction tests are those tests which are accomplished on units selected as specified in 4.1.2 and are for the purpose of determining

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that the production methods are capable of providing items that comply with the technical requirements of the contract.

4.4 Test equipment.- The following tems of test equipment are required to perform the acceptance tests set forth in this specification.

4.4.1 Transmitter, capable of providing a 60 ± 2 kc trigger pulse with a peak to peak amplitude of 760 ± 40 volts, a pulse length of two plus or minus 0.5 ms and having a pulse repetition rate of 96 ± 6 pulses per minute.

4.4.2 Time interval meter, capable of measuring time intervals of from zero to 4000 ms with an accuracy of at least one percent.

4.4.3 Microvoltmeter, capable of measuring two uv rms to 100 volts rms with an accuracy of at least two percent.

4.4.4 Oscilloscope (Dumont 304 H), ZX 16-078495-3801 or equal.

4.4.5 Basic Unit assembly per BuOrd Drawing 1473729.

4.4.6 Power Source, 115 ± 2 volts ac, 60 ± 6 cps and capable of supplying a resistive load three amperes of current.

4.4.7 Vacuum tube voltmeter, capable of measuring zero to 1000 volts ac or dc with an accuracy of at least three percent. The ac frequency range shall be at least 20 cps to 100 kc. The input impedance shall be at least ten megohms.

4.5 Acceptance tests

4.5.1 Except where laboratory tests are indicated, the contractor shall conduct acceptance tests as specified in 4.3.1, to assure that the plug in unit is in compliance with the requirements of 3.2, 3.4, 3.6 and 3.7.

4.5.2 Test conditions.- Unless otherwise specified, the plug in unit shall be subjected to acceptance tests under the following conditions.

4.5.2.1 Temperature:	Room ambient 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C)
4.5.2.2 Altitude:	Normal ground atmospheric pressure
4.5.2.3 Vibration:	None
4.5.2.4 Humidity:	Room ambient to 95 percent relative maximum

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4.5.3 Test and inspection equipment and facilities.- The manufacturer shall furnish and maintain all necessary test equipment and facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.4 Test procedure.

4.5.4.1 Set the Basic Unit as follows:
Pulse - CW switch to Pulse position,
Range - Sel. switch to position one
DB dials to zero db total.
Power switch to the off position.

4.5.4.2 Set the Plug In Unit as follows:
Frequency - Select to 60.0 kc
Up down zero switch to zero
Set oscilloscope Switch to Pin g

4.5.4.3 Supply the Basic Unit with 115 volt ac power from the power source (4.4.6), turn the power switch on and allow the unit to warm up for at least one half hour.

4.5.4.4 B-plus voltage.- Using the VTVM (4.4.7) measure the B-plus voltage at J2 on the unit chassis. The voltage shall meet the requirement of 3.4.2.

4.5.4.5 Output echo signal.- Energize the transmitter (4.4.1) and apply trigger pulses as per 3.4.3 to J-10 pin C on the Plug In Unit.

4.5.4.5.1 Adjust the oscilloscope (4.4.4) until a clear presentation of the echo signal pulse is obtained. The pulse shall meet the requirement of 3.4.4.6 and 3.4.4.1.

4.5.4.6 Time delay.- Connect the time interval meter (4.4.2) to the range jack of the Basic Unit and measure the time delay from the leading edge of the trigger pulse to the leading edge of the echo signal pulse. The unit shall perform as follows to meet the requirement of 3.4.4.2.

4.5.4.6.1 Check target range simulation by switching the Range Select Switch (S-3) through the following positions and observing the time interval meter .(4.4.2) for proper echo delay time:

Switch Position	Echo Time Delay (Milliseconds)
1	80 ± 2%
3	480 ± 2%

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4.5.4.7 Pulse length.- Connect the time interval meter to the pulse length (PL) jack on the Basic Unit and measure the pulse length of the echo signal pulse. The Target Echo Pulse Lengths hall be 35 ± 20% milliseconds.

4.5.4.8 Noise level.- Turn off the transmitter and set the unit for pulse operation. Connect the Micro-Voltmeter (4.4.3) to pin E on J10. With the Up Down zero switch in up the output voltage (noise) shall measure less than ten uv rms to meet the requirement of 3.4.4.5: Record the voltage.

4.5.4.9 Output voltage.- Set the unit for CW operation and measure the output on pin E of J 10 of the Plug In Unit for db attenuation settings of zero, ten, 20, 40, 60 and 80, Record the voltage at each setting and reduce the value by the value of the recorded noise level (4.5.4.8). The output voltage shall meet the requirement of 3.4.4.4.

4.5.4.10 High level output.- Connect the vacuum tube voltmeter (4.4.7) to pin A on J9 of the Plug In Unit. With the unit set for CW operation, the high level output voltage shall be at least seven volts rms to meet the requirement of 3.4.5.

4.5.4.11 Manual operation.- Set the unit for pulse operation and depress and release the manual trigger switch on the unit. A pulse shall appear on the oscilloscope each time the switch is depressed to meet the requirement of 3.4.6.

4.5.4.12 Frequency check.- Connect the T I and E put meter to pin A on the Scope jack (J-9) and check frequency at 60.0 kc and 60.4 kc positions of the Frequency Select Switch. The measured values, in each case, shall be within 50 cps of the nominal.

4.5.4.13 Phase shift.- To check the phase shift, connect the Microvoltmeter (4.4.3) to J-10, the low level output receptacle, as follows:

Pin D to the case or ground side of the meter
Pin E to the high side of the meter

Set the pulse CW switch to CW, and the frequency select switch to 60.4 kc. With the Up-Down zero switch in the zero position and the attenuator on the Basic Unit set on zero db, the output adjust potentiometer on the Basic unit should be set such that the meter reads exactly 0.01 volt. With the settings as above, the microvoltmeter should now be connected wuch that pin C of J 10 is connected to the ground side (case) of the meter and pin E of J 10 to the high side of the meter. With this connection, the microvolt meter should read as follows for given settings on the Up-Down zero switch.

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Up Down Zero Switch set

Zero (corresponds to 6 ± 2° phase shift)	698 to 1395 microvolts
Down (corresponds to 25.5 ± 3° phase shift)	3.9 to 4.9 millivolts
Up (corresponds to 25.5 ± 3° phase shift)	3.9 to 4.9 millivolts

4.5.5 The tests of 4.5,5.1 through 4.5.5.5 shall be performed as follows:

Step 1. One unit from each of the first three groups of five manufactured shall be subjected to the tests of 4.5.5.1 through 4.5.5.5 by a laboratory. designated by the procuring activity. Failure of any sample unit will be cause for rejection of the group represented.

Step 2. Upon satisfactory compliance of three consecutive sample units with the tests of 4.5.5.1 through 4.5.5.5, inspection for the remainder of the contract shall be based on MIL-STD-105, inspection L4, acceptance number zero.

Step 3. When a defective unit is found while the L4 level of inspection is in effect, shift to step 1 and proceed as indicated.

4.5.5.1 Vibration, nonoperating condition.- The unit shall be subjected to vibration tests so as to meet the requirements specified in 3.5.1 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification

4.5.5.1.1 Procedure.- The unit shall be mounted rigidly to the vibration testing device and vibrated according to Schedule I in each of the unit axes (3.3.5).

SCHEDULE I

Frequency Range	Amplitude	Minimum test duration		
		Room temp.	-20°F	+130°F
10 - 30 cps	0.060 ± 0.006 in	60 min	15 min	15 min
31 - 60 cps	3g ± 0.3g	60 min	15 min	15 min
61 - 100 cps	3g ± 0.3g	60 min	15 min	15 min

The frequency shall be continuously varied between each of the limits given, at a uniform rate over a three to five minute period for the duration of the test.

As an alternate method the unit shall be vibrated at specific frequencies differing by not more than five cps in each of the unit axes and covering the entire frequency

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range and amplitude of Schedule I. The total test time shall not be less than that of Schedule I and the test times at the specific frequencies shall be equally divided.

4.5.5.2 Shock, ready nonoperating condition.- The unit shall be subjected to shock tests so as to meet the requirements specified in 3.5.2 and thereafter meet all the performance requirements and product characteristics of this specification.

4.5.5.2.1 Procedure.- The unit shall be mounted rigidly to the shock testing device as outlined in MIL-T-945. Apply three hammer blows parallel to each of the three principal axes of the unit. The height of the hammer drop on each side shall be one foot, one and one half feet, and two feet.

4.5.5.3 Shipping condition tests

4.5.5.3.1 Shock-transportation, storage and handling. The unit, packaged and packed as specified in section 5 shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

4.5.5.3.2 Vibration-transportation. The unit, packaged and packed as specified in section 5, shall be vibration tested as specified in Specification MIL-W-21927.

4.5.5.3.3 Functional tests. At the conclusion of the tests specified in 4.5.5.3.1 and 4.5.5.3.2 above, functional and other tests as applicable, shall be conducted on the unit to determine freedom from operational malfunction and compliance with all requirements of section 3.

4.5.5.3.4 Waive of tests. Tests specified in 4.5.5.3.1 and 4.5.5.3.2 above, may be waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.5.5.4 Temperature.- The unit shall be subjected to temperature tests so as to meet the requirements specified in 3.5.4.

4.5.5.5 Operating life.- Operate the unit continuously for 720 hours while triggering the unit at least once per second. At the end of each 24 hour period test the unit as per 4.5.4. The unit shall pass these tests to meet the requirement of 3.5.5.

4.6 Rejections.- Failure of the material under consideration to meet any of the requirements and tests of this specification shall be considered cause for rejection of the lot represented and units tested.

5. PREPARATION FOR DELIVERY

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5.1 Preservation and packaging.

5.1.1 Level A. In addition to the following Level A packaging shall be in accordance with MIL-E-17555.

5.1.1.1 Cleaning, drying and preservative application. Nonapplicable

5.1.1.2 Unit packaging. Unit packaging of the unit shall be in accordance with Method II e of Specification MIL-P-116. The unit package shall contain one unit.

5.1.1.3 Intermediate package. Nonapplicable

5.1.1.4 Cushioning. The unit package and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to ensure that the unit will meet all the requirements of section 3 after being tested in accordance with 4.5.5.3.1 and 4.5.5.3.2.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than five percent. The PH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. Not applicable

5.1.3 Level C

5.1.3.1 Cleaning, drying and preservative application. Not applicable

5.1.3.2 Unit packaging. Unit packaging shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one unit.

5.1.3.3 Intermediate package, Nonapplicable.

5.2 Packing

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with Specification MIL-E-17555.

5.2.1.1 Exterior containers. Units packaged as described in 5.1.1.2 shall be packed in a fiberboard box in accordance with PPP-B-636 Specification Type I class 2 or Type II Class 2 (gross weight not to exceed 70 lbs.)

5.2.2 Level B. In addition to the following, Level B, packing shall be in accordance with Specification MIL-E-17555.

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5.2.2.1 Exterior containers. Units packaged as described in 5.1.1.2 shall be packed in a fiberboard box in accordance with Specification PPP-B-636 Type I Class I or Type II Class I (gross weight not to exceed 70 lbs).

5.2.3 Level C. Units packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special markings. None, unless otherwise specified.

5.3.2 Normal markings. In addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers,

6. NOTES

6.1 Intended use.- The Plug In Unit is intended for use with Test Set Mark 315 Mod 1 when testing Torpedo Mark 32 Mod 2 acoustic panel.

6.2 Ordering data

6.2.1 Procurement documents should containe the specific title, number and date of this specification, applicable drawings and other documents.

6.2.2 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116, and to paragraphs 4.5.5.3.4 and 5.3.2.

6.2.3 Criteria for use of proper level of preservation, packaging and packing shall be as follows.

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

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For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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MIL-P-23108(Wep)
Amendment 1
9 August 1963

DOWNGRADED BY AUTHORITY OF
NAVORDINST 5511.35 ORD-055
DATED 2 MAY 1973

MILITARY SPECIFICATION

PLUG-IN UNIT FOR TEST SET, ECHO SIMULATOR (U)

This amendment forms a part of Military Specification MIL-P-23108(Wep) of 30 Nov 1961, and has been approved by the Bureau of Naval Weapons, Department of the Navy

Page 2, under "PUBLICATIONS": Delete "OP400" and associated title.

Page 4, paragraph 3.2.7. Delete and substitute:
"3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings."

Page 4, paragraph 3.2.8: Delete.

Page 10, paragraph 4.5.4.4: Delete "meet the requirement of 3.4.2" and substitute "be 300 plus or minus 10 volts dc".

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